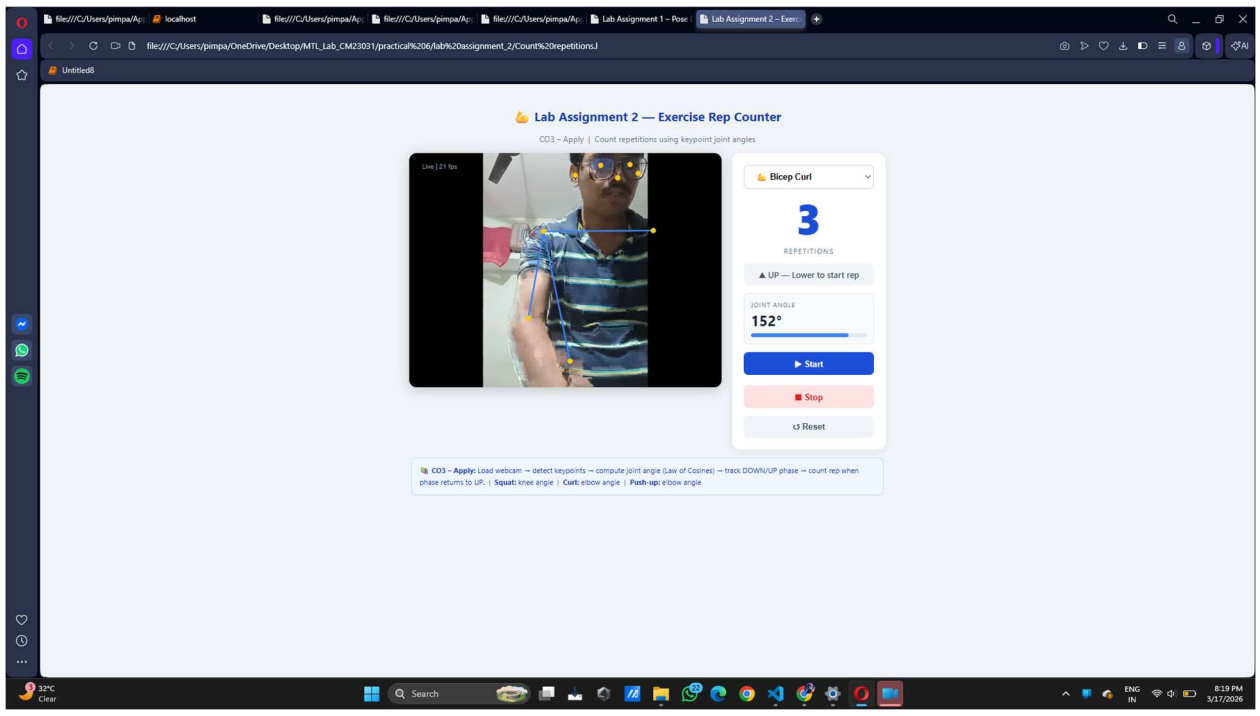
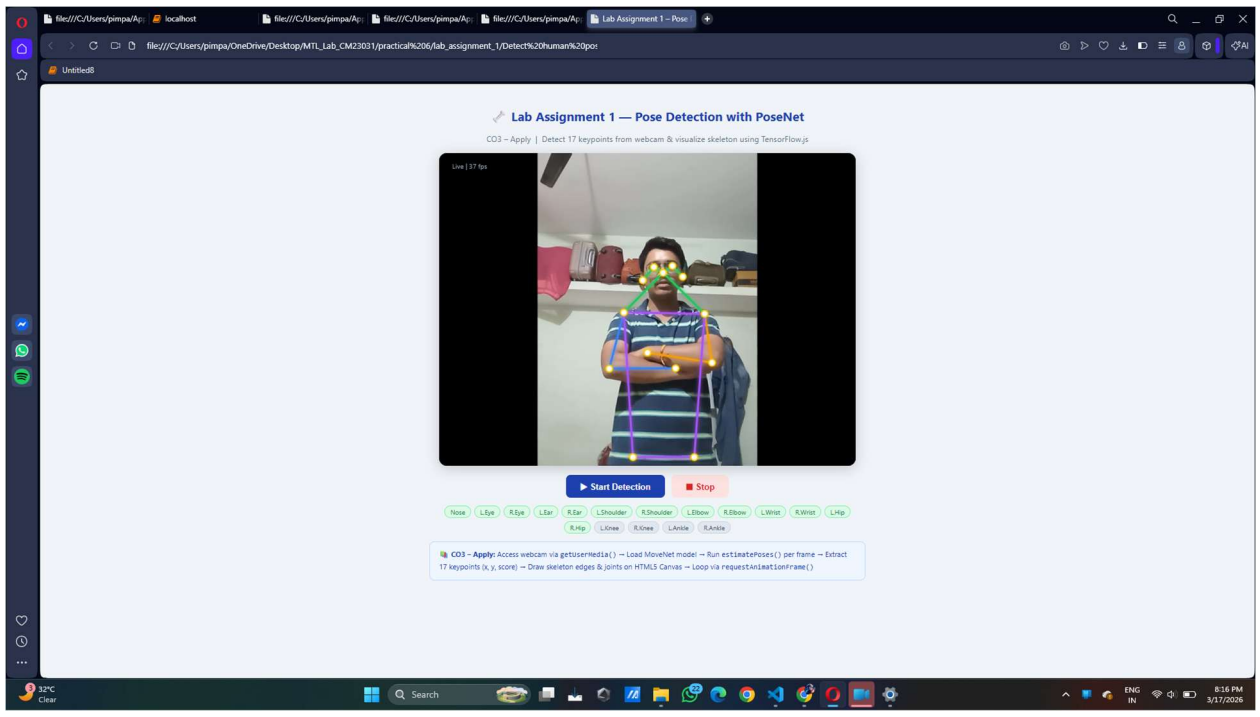
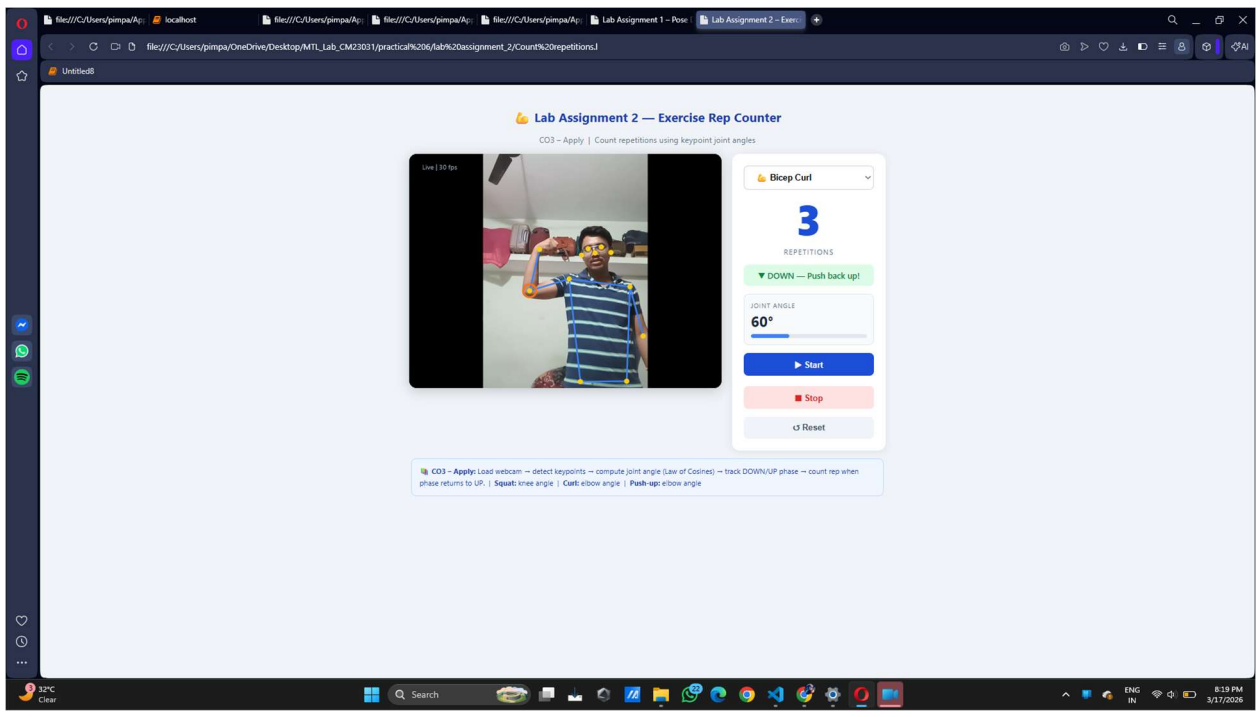
Practical 6

assignment output:1

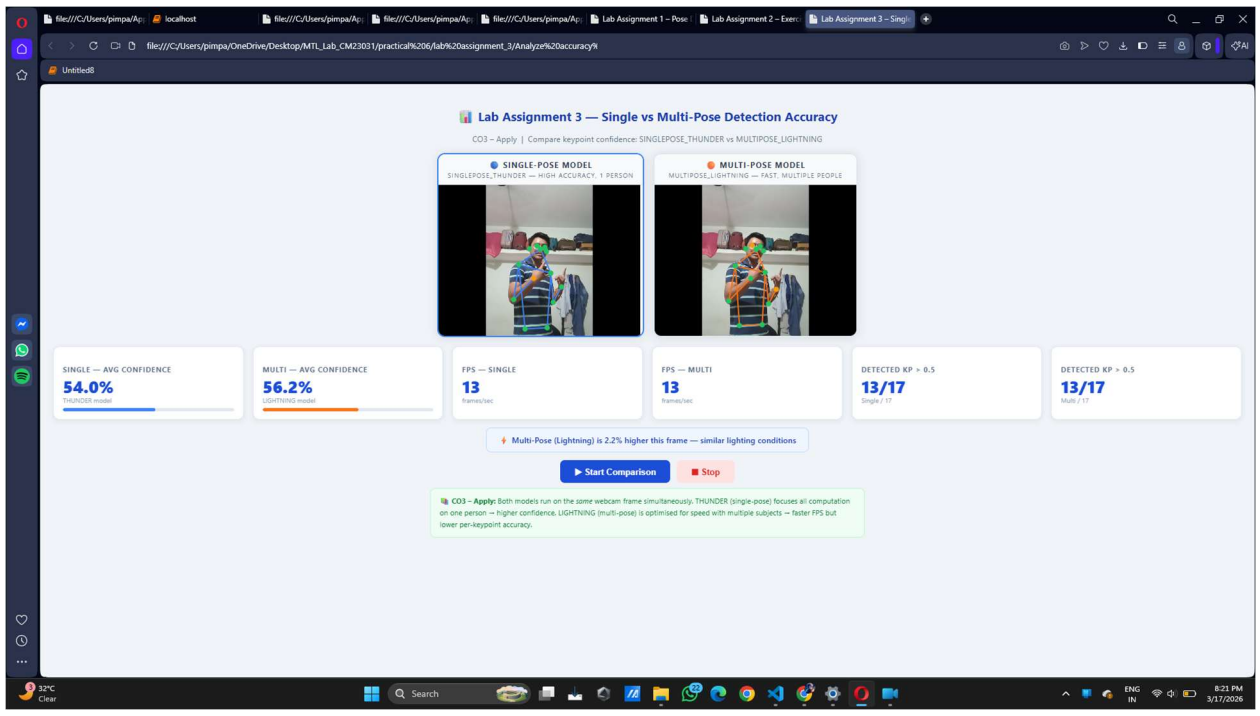


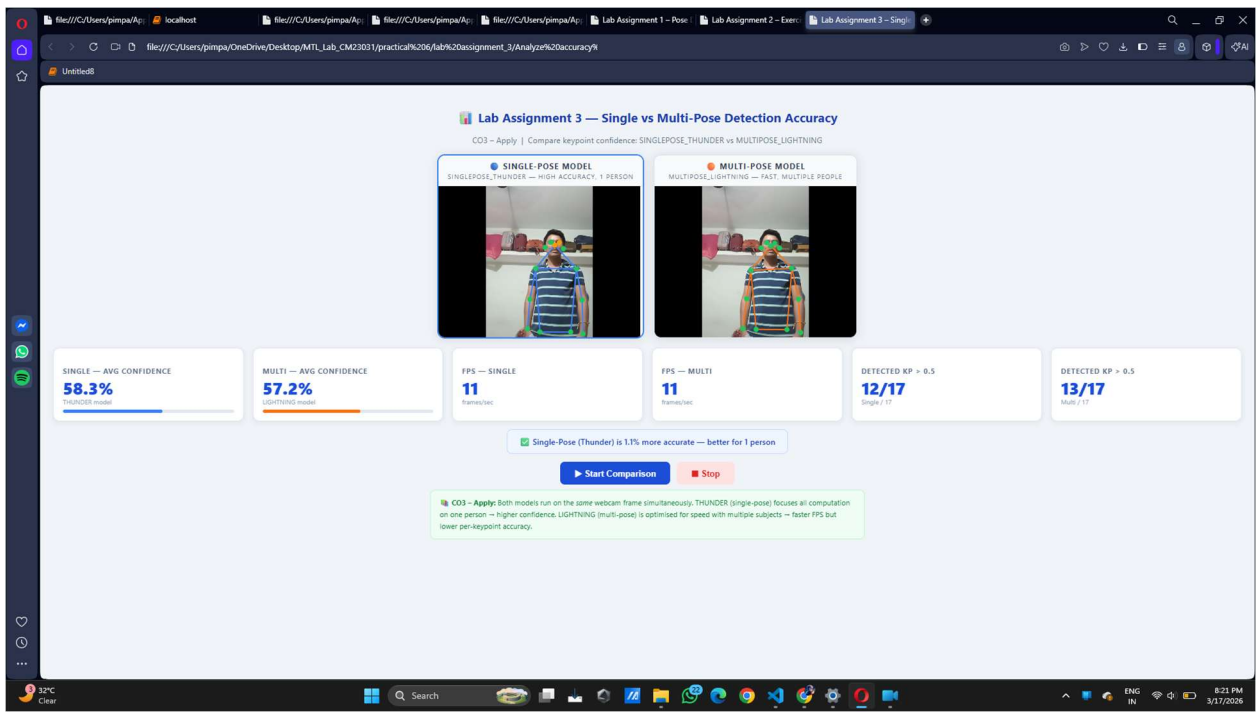


LAB ASSIGMENT 2:



LAB ASSIGMNET 3:


Lab Assignment 3 — Single vs Multi-Pose Detection Accuracy
CO3 – Apply | Compare keypoint confidence: SINGLEPOSE_THUNDER vs MULTIPOSE_LIGHTNING
SINGLE-POSE MODEL
SINGLEPOSE_THUNDER — HIGH ACCURACY, 1 PERSON
MULTI-POSE MODEL
MULTIPOSE_LIGHTNING — FAST, MULTIPLE PEOPLE
SINGLE — AVG CONFIDENCE
54.0%
THUNDER model
MULTI — AVG CONFIDENCE
56.2%
LIGHTNING model
FPS — SINGLE
13
frames/sec
FPS — MULTI
13
frames/sec
DETECTED KP > 0.5
13/17
Single / 17
DETECTED KP > 0.5
13/17
Multi / 17
Multi-Pose (Lightning) is 2.2% higher this frame — similar lighting conditions
Start Comparison
Stop
CO3 – Apply: Both models run on the same webcam frame simultaneously. THUNDER (single-pose) focuses all computation on one person → higher confidence. LIGHTNING (multi-pose) is optimised for speed with multiple subjects → faster FPS but lower per-keypoint accuracy.


Lab Assignment 3 — Single vs Multi-Pose Detection Accuracy
CO3 – Apply | Compare keypoint confidence: SINGLEPOSE_THUNDER vs MULTIPOSE_LIGHTNING
SINGLE-POSE MODEL
SINGLEPOSE_THUNDER — HIGH ACCURACY, 1 PERSON
MULTI-POSE MODEL
MULTIPOSE_LIGHTNING — FAST, MULTIPLE PEOPLE
SINGLE — AVG CONFIDENCE
58.3%
THUNDER model
MULTI — AVG CONFIDENCE
57.2%
LIGHTNING model
FPS — SINGLE
11
frames/sec
FPS — MULTI
11
frames/sec
DETECTED KP > 0.5
12/17
Single / 17
DETECTED KP > 0.5
13/17
Multi / 17
Single-Pose (Thunder) is 1.1% more accurate — better for 1 person
Start Comparison
Stop
CO3 – Apply: Both models run on the same webcam frame simultaneously. THUNDER (single-pose) focuses all computation on one person → higher confidence. LIGHTNING (multi-pose) is optimised for speed with multiple subjects → faster FPS but lower per-keypoint accuracy.